

Yene Irvine

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Education & Certifications

M.ENG., SOFTWARE ENGINEERING

University of Calgary, Calgary AB
GPA of 3.94; graduated May 2024

B.ENG., CHEMICAL ENGINEERING

Dalhousie University, Halifax, NS
GPA of 3.80; graduated April 2018

AWS CERTIFIED CLOUD PRACTITIONER CLF-C02, 2024

MICROSOFT CLOUD CERTIFIED: AZURE FUNDAMENTALS AZ-900, 2024

Work Experience

DATA ENGINEER | Fuel Transport Inc. | Montreal, QC | September 2024 – Present

- Built and integrated a Snowflake Rate Editor (FastAPI, React) to enable real-time editing of client rates, powering 100s of daily API calls and contributing to \$1.5M+ in revenue over 5 months
- Developed and implemented a custom Azure Data Factory pipeline to migrate data from IBM DB2 to Snowflake, replacing Fivetran (a managed ETL platform) and saving \$40K+ annually in third-party costs
- Optimized CI/CD pipelines (GitHub Actions, Docker, Azure Container Apps) to streamline deployment of the Snowflake Rate Editor's frontend and backend containers, improving reliability and reducing overhead
- Leveraged Azure Monitor, Application Insights, and related Azure services to track API performance, analyze traffic patterns, and proactively resolve issues, ensuring high availability for enterprise clients

CONTROLS ENGINEER | MCA Consultants | Charlottetown, PE & Halifax, NS | Apr. 2019 – Jan. 2022

Worked on 2 large-scale expansion projects w/ a pharmaceutical manufacturing company. My key contributions:

- Developed and commissioned 25 complex control narratives for utilities and process control systems, collaborating closely with PLCs, process equipment, contractors to enhance BV Building Automation System
- Developed a program integrated into Microsoft Excel to determine relieving requirements for pressurized vessels based on NFPA and ASME standards; this program was integral in designing venting configurations and specifying required relieving equipment for 50+ major pressurized process vessels
- Led a team to commission all process and utility systems and associated equipment under budget

Relevant Projects

- **Detecting Breathing Irregularities with Deep Learning:** Developed and deployed a 3D Convolutional Neural Network with University of Calgary Biometrics Lab to detect sleep-related medical emergencies via thermal video, leveraging HPC compute and WandB for real-time tracking at the Calgary Drop-In Centre
- **RunIt:** a mobile running app using JavaScript (React Native) and NestJS, integrating real-time GPS tracking, Firebase authentication, and social features to enhance user engagement
- **AI DJ Setlister:** a cloud-based web application that leverages Spotify's web API and custom-built optimization algorithms to dynamically generate and optimize DIY-DJ set playlists, hosted on AWS EC2

Technical Skills & Languages

- **Frontend, Backend Development:** Node.js, FastAPI, TypeScript, JavaScript (ReactJS), API development, SQL
- **Cloud & DevOps:** AWS (Certified), Azure (Certified, incl. Functions, Container Apps, ADF, Monitor, Application Insights), GCP, Docker, Terraform, GitHub Actions, CI/CD
- **Data & ML:** ETL/ELT pipelines, Data Warehousing (Snowflake, DB2), Python (Pandas, NumPy, PyTorch), ML model development & tuning & deployment, HPC compute, Weights & Biases (WandB)